

COMPASS with CRE-to-gene annotations for Alzheimer disease

July 2026

Abstract

COMPASS estimates gene-by-context mediated trait contributions from GWAS summary statistics, SNP-to-gene annotations, and linkage disequilibrium (LD). An initial implementation selected the nuclear-norm penalty by leaving out entire chromosomes. That validation design is inappropriate for a gene-indexed model because most held-out chromosomes contain genes that are absent from the training chromosomes; it therefore tests extrapolation to unseen genes rather than generalization across regulatory elements for the same gene. We replaced chromosome-heldout validation with a CRE-structured design and switched the default annotations from one top eQTL per gene and cell type to public Activity-by-Contact (ABC) enhancer-gene scores. The revised workflow holds out LD-separated candidate regulatory element (CRE) groups within genes, so the same gene can contribute independent regulatory elements to both training and validation folds.

1 Model

Let χ_j^2 be the marginal GWAS statistic for variant j , and let $a_{jkl} \geq 0$ denote the annotation linking variant j to gene k in context l . In the revised implementation, a_{jkl} is derived from CRE-to-gene ABC scores: a GWAS variant receives the ABC score of a CRE-gene pair when the variant lies inside that CRE. The context index l denotes the ABC biosample or cellular context, with an optional intercept context.

LD-smoothed annotations are

$$T_{jkl} = \sum_m r_{jm}^2 a_{mkl}, \quad (1)$$

where r_{jm}^2 is squared LD. COMPASS models the expected summary statistic as

$$E[\chi_j^2] = 1 + N_j \left(\sum_{k,l} B_{kl} T_{jkl} + \tau l_j \right), \quad (2)$$

where $B_{kl} \geq 0$ is the mediated contribution of gene k through context l , $\tau \geq 0$ is a residual polygenic component, and $l_j = \sum_m r_{jm}^2$ is the LD score. The fitted objective is

$$\min_{B \geq 0, \tau \geq 0} \frac{1}{J} \sum_j w_j \left\{ \chi_j^2 - 1 - N_j \left(\sum_{k,l} B_{kl} T_{jkl} + \tau l_j \right) \right\}^2 + \lambda \|B\|_*. \quad (3)$$

The nuclear norm encourages shared low-rank gene-by-context structure while the non-negativity constraint prevents LD-correlated neighboring annotations from fitting the same signal through cancelling positive and negative coefficients.

2 ABC annotations

The default annotation source is the public Nasser et al. ABC prediction table: all element-gene connections with ABC score at least 0.015 across 131 biosamples. The ABC model predicts enhancer-gene regulation from enhancer activity and enhancer-promoter contact; the public table is distributed by the Engreitz Lab and is linked from their resource page. The ABC documentation defines the score as activity times contact, normalized over candidate elements within 5 Mb of the gene promoter, and notes that predictions are same-chromosome CRE-gene links.

For Alzheimer disease (AD), the default run uses the brain-labelled biosamples available in this table: `astrocyte-ENCODE`, `bipolar_neuron_from_iPSC-ENCODE`, and `H1_Derived_Neuronal_Progenitor_Cultures`. Because the public table does not contain a microglia biosample, the default also includes myeloid proxy contexts: `CD14-positive_monocyte-ENCODE`, `CD14-positive_monocytes-Roadmap`, and `THP-1_macrophage`. The full 131-biosample table is downloaded and can be selected by passing `--abc-cell-types all`.

The implementation reads the ABC table in chunks, intersects each CRE interval with GWAS variants, and emits sparse variant-gene-context triples whose values are ABC scores. The regression row universe is not restricted to ABC-overlapping variants. Instead, COMPASS uses all GWAS variants represented in the UK Biobank LD reference panel. Variants outside ABC CREs have all-zero annotation rows but remain in the regression so they contribute to LD tagging and the residual non-ABC-mediated LD-score term. Nearby CREs for the same gene are collapsed into coarse LD-distance clusters before fold assignment.

LD is represented as chromosome-level sparse blocks rather than a single genome-wide matrix. Raw UK Biobank correlations are converted to r^2 , entries with $r^2 < 0.01$ are discarded, and the retained LD values are stored and applied as half-precision sparse tensors during fitting. Other model tensors default to single precision, with a command-line option to compare half-precision model tensors against the default while keeping LD half precision in both runs.

3 Why chromosome-heldout CV was removed

Leaving out an entire chromosome is not a valid penalty-selection scheme for the gene-indexed COMPASS parameter B . If chromosome 1 is held out, then most genes on chromosome 1 have no training annotations. Prediction on the held-out chromosome therefore relies on gene effects that were not estimable in the training set. This design can only evaluate whether a globally shared context profile transfers to unseen genes; it cannot evaluate whether estimated gene-level contributions generalize to independent regulatory elements for the same gene. The previously reported chromosome-heldout null result should therefore be treated as a diagnostic of an invalid validation split, not as an AD biological conclusion.

4 CRE-structured CV

The revised validation design assigns fold labels to CRE groups rather than chromosomes. For each gene, CREs are sorted by genomic position after grouping nearby CREs into LD-distance clusters. Independent clusters for the same gene are assigned across folds. Variant-level fold labels are then inherited from the CRE groups they overlap; variants overlapping multiple conflicting folds are retained for training but are not used as validation rows. For a validation fold, the model is trained on all variants not assigned to that fold and evaluated on variants assigned to the held-out CRE fold.

This design has the intended target: the model learns B_{kl} from some regulatory elements for a gene and tests whether those estimates predict other, approximately LD-independent regulatory elements for the same gene. It also preserves genome-wide LD modeling because the held-out units are local CRE clusters rather than whole chromosomes.

5 Implementation status

The repository now downloads the public ABC table into `raw/abc/AllPredictions.AvgHiC.ABC0.015.minus150.` and uses ABC annotations by default in `scripts/run.py`. Dataset caches store CRE fold labels alongside GWAS statistics, chromosome labels, and sample sizes. The row set is filtered to variants present in the UK Biobank LD metadata, and non-CRE rows are retained with zero annotations. The previous leave-one-chromosome CV code path has been removed from the fitting path; requesting CV now requires CRE fold labels in the dataset. The older top-eQTL annotation loader remains available for debugging, but it should be run with `--no-cv` unless a valid CRE fold structure is added.

6 Next analysis

The corrected AD analysis should be interpreted only after fitting the ABC-based design with CRE-structured lambda selection. The primary outputs will be global gene scores, biosample-specific gene scores, and ordered gene-set enrichment using the assayed genes as background. Any enrichment should be reported only from the CRE-validated lambda-selected fit; weaker penalties can be shown as sensitivity analyses but should not define discoveries.